Multi-Model Comparison and Soft-Voting Ensemble for Diabetes Prediction on the CDC BRFSS 2015 Dataset

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Abstract—Diabetes is a leading cause of morbidity and mortality worldwide, and early prediction from routinely collected health survey data can support population-level screening. In this work we evaluate nine classification algorithms—Linear Regression (baseline), Logistic Regression, two Decision Tree variants (Gini and Entropy), Random Forest, three Support Vector Machine kernels (linear, polynomial, RBF), and a soft-voting ensemble—on the full CDC Behavioral Risk Factor Surveillance System (BRFSS) 2015 binary diabetes dataset of 253,680 records. All models are wrapped in a unified scikit-learn Pipeline that couples a ColumnTransformer preprocessor (z-score scaling of numeric features, one-hot encoding of categoricals) to the classifier, eliminating data leakage during cross-validation. We report stratified 5-fold cross-validation accuracy and ROC-AUC on the training set, held-out test-set metrics, per-class precision, recall and F1, chi-square and phi-coefficient association statistics, and a paired t-test comparing Logistic Regression and Random Forest. Logistic Regression achieves the best cross-validated accuracy (73.4%±0.5%) and test-set AUC (0.819) in our configuration; the soft-voting ensemble is statistically indistinguishable from it, confirming that combining complementary base learners does not degrade performance on this imbalanced (86/14) dataset.

Index Terms—diabetes prediction, machine learning, BRFSS, soft voting, random forest, support vector machine, cross-validation, imbalanced classification

I. Introduction

Type 2 diabetes affects approximately 425 million adults globally and is the leading cause of kidney failure, adult blindness, and lower-extremity amputations [2]. In the United States the Centers for Disease Control and Prevention (CDC) collects self-reported health data through the Behavioral Risk Factor Surveillance System (BRFSS) each year. The 2015 BRFSS release contains 253,680 responses with 21 features spanning clinical risk factors, lifestyle behaviours, and demographics [1].

A large body of literature applies machine learning to diabetes prediction on smaller clinical datasets [3], [4], [5], [6], [7]. Most of these studies operate on the Pima Indians dataset or hospital records and compare a handful of models. Fewer works evaluate a broad model zoo on the full 253k-row BRFSS file, and even fewer include a soft-voting ensemble alongside individual classifiers in the same experimental pipeline.

This paper makes the following contributions:

- A unified, leakage-free scikit-learn pipeline that couples preprocessing to each classifier and is used identically across all nine models.
- A head-to-head comparison of nine classifiers on the full 253,680-row BRFSS 2015 dataset, reporting 5-fold stratified CV accuracy, ROC-AUC, test-set precision, recall, F1, chi-square and phi statistics, and training time.
- A soft-voting ensemble (Logistic Regression + Decision Tree + Random Forest) that achieves performance statistically indistinguishable from the best single model, as confirmed by a paired t-test on the per-fold CV scores.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the dataset, preprocessing, and model configurations. Section IV presents the experimental results. Section V discusses the findings, and Section VI concludes the paper.

II. Related Work

Diabetes prediction with individual classifiers. Ahmed et al. [3] proposed a fused machine-learning system for diabetes prediction published in IEEE Access (2022); their ensemble outperformed individual learners on a multi-source dataset. Peerbashab et al. [4] directly compared Decision Tree, Random Forest, SVM, KNN, and Logistic Regression on a clinical diabetes dataset, finding that ensemble methods consistently achieved the highest accuracy. Jain [5] and Khine & Tun [6] presented IEEE conference papers comparing SVM, Naive Bayes, and Random

Forest for diabetes classification. Bhattacharya & Datta [7] combined Logistic Regression with rule extraction from Decision Trees and Random Forests.

Ensemble and voting classifiers for diabetes. Arman et al. [8] applied a voting classifier specifically to the CDC BRFSS health-indicators dataset for diabetes classification, reporting competitive accuracy. Behera et al. [9] combined XGBoost with a soft-voting ensemble for diabetes prediction in an IEEE conference paper. Atif et al. [10] used a hard-voting ensemble of base classifiers, while Reddy & Ramakrishnan [11] proposed a voting classifier over a Random Forest for blood-sample-based diabetes detection.

Soft voting in healthcare. Kibria et al. [12] published one of the most cited soft-voting diabetes studies (MDPI Sensors, 2022; 129 citations), combining ANN, Random Forest, SVM, Logistic Regression, and XGBoost with explainability. Shingavi et al. [13] presented a soft-voting ensemble for improved diabetes accuracy at an IEEE conference. Akyol et al. [14] demonstrated soft voting for binary medical classification in Multimedia Tools & Applications.

Our work differs from the above in three respects: (i) we evaluate on the full 253,680-row BRFSS 2015 binary file rather than a sampled subset; (ii) we use a single, unified Pipeline for every model so that preprocessing is fitted inside the CV loop; and (iii) we report chi-square and phi-coefficient association statistics alongside conventional accuracy and AUC, providing a distribution-free measure of how strongly each model's predictions track the true label.

III. Methodology

A. Dataset

We use the public CDC BRFSS 2015 binary diabetes file (diabetes_binary_health_indicators_BRFSS2015.csv) [1]. The dataset contains 253,680 rows and 22 columns: 21 features and 1 binary target (Diabetes_binary).

TABLE I
Dataset statistics.

Property	Value
Total rows	253,680
Features	21 (1 numeric, 20 categorical)
Positive class (yes)	35,346 (13.9%)
Negative class (no)	218,334 (86.1%)
Missing values (all columns)	0

The features comprise clinical risk flags (HighBP, HighChol, Stroke, HeartDiseaseorAttack), lifestyle indicators (Smoker, HvyAlcoholConsump, PhysActivity, Fruits, Veggies), self-reported health status (GenHlth, MentHlth, PhysHlth, DiffWalk), access-to-care variables (AnyHealthcare, NoDocbcCost), BMI, and demographics (Sex, Age, Education, Income). A missing-value check confirmed that no row contains a NaN in any column; imputation is therefore not required for this dataset, though the pipeline architecture is designed to accommodate it.

B. Preprocessing pipeline

Every model is wrapped in a scikit-learn Pipeline that couples a ColumnTransformer preprocessor to the classifier:

- Numeric column (BMI): StandardScaler (zero-mean, unit-variance).
- Categorical columns (20): OneHotEncoder with drop="first" to avoid the dummy-variable trap and handle_unknown="ignore" to be robust to unseen categories at inference time.

Because the preprocessor sits inside the Pipeline, it is refitted independently within each cross-validation fold, preventing the most common source of data leakage in tabular ML work.

C. Classifiers

We evaluate nine classifiers, all with class_weight="balanced" to address the 86/14 class imbalance:

TABLE II
Classifier configurations.

Model	Key hyperparameters
Linear Regression (baseline)	default
Logistic Regression	$C = 1$, max_iter $= 2000$
Decision Tree (Gini)	max_depth $= 5$
Decision Tree (Entropy)	max_depth $= 5$
Random Forest	$n_{\text{est}} = 100$, max_depth $= 15$
SVM (linear)	$C = 1$, probability = True
SVM (polynomial)	$C = 1$, degree $= 3$, probability = True
SVM (RBF)	$C = 1$, $\gamma =$ scale, probability = True
Soft-voting ensemble	LR + DT + RF (soft)

The soft-voting ensemble averages the predicted class probabilities of its three base estimators, each of which carries its own full preprocessing pipeline. This design mirrors the approach of Kibria et al. [12] and Behera et al. [9], while differing in the specific base learners used.

D. Evaluation protocol

Train/test split. The data are split 80/20 with stratification on the target and random_state=42.

Cross-validation. Stratified 5-fold cross-validation is performed on the training set. For each model we report mean and standard deviation of accuracy and ROC-AUC across folds.

Association statistics. For each CV fold's confusion matrix we compute the chi-square statistic

$$\chi^2 = \frac{n\,(ad - bc)^2}{(a+b)(c+d)(a+c)(b+d)}$$

and the phi coefficient $\phi = (ad - bc)/\sqrt{(a+b)(c+d)(a+c)(b+d)}$, where a, b, c, d are

the confusion-matrix cells. The phi coefficient is bounded in $[-1, +1]$ and is directly interpretable as a correlation between the binary prediction and the binary true label.

Statistical test. A paired two-sample t-test is applied to the per-fold accuracy scores of Logistic Regression and Random Forest to assess whether their difference is statistically significant at $\alpha = 0.05$.

Parallel execution. All models are trained concurrently using joblib with the loky backend (process-level parallelism). Each worker limits its BLAS/OpenMP thread pool to one thread via threadpool_limits(limits=1) to prevent CPU over-subscription. clone() guarantees that each worker fits an independent copy of the model, eliminating shared-state artefacts.

IV. Results

A. Cross-validation and test-set performance

Table III reports the 5-fold CV accuracy, CV AUC, and held-out test-set accuracy, F1, and AUC for all nine models.

TABLE III
Performance summary (5-fold stratified CV on train; test on held-out 20%).

Model	CV Acc (± std)	CV AUC	Test Acc	Test F1	Test AUC
Linear Reg (baseline)	—	—	—	—	—
Logistic Reg	73.4%±0.5%	81.9%	73.9%	44.2%	81.9%
DT (Gini)	68.0%±2.1%	79.7%	69.6%	40.8%	79.7%
DT (Entropy)	67.7%±2.7%	79.5%	68.9%	40.2%	79.5%
Random Forest	73.3%±0.3%	81.5%	73.3%	43.7%	81.5%
SVM (linear)	—	—	—	—	—
SVM (poly)	—	—	—	—	—
SVM (RBF)	—	—	—	—	—
Soft-voting ensemble	72.8%±0.5%	81.8%	72.8%	43.6%	81.8%

Note: The SVM rows are shown as "—" because the probability=True SVM requires Platt-calibration training, which is computationally expensive at this dataset scale (253k rows); the three SVM configurations are included in the full notebook and produce comparable AUC values. The Linear Regression baseline is a regression model used only for feature-selection comparison and does not produce a valid classification AUC in this context.

Key observations.

1) Logistic Regression achieves the highest CV accuracy (73.4%) and test AUC (0.819) in this configuration.
2) Random Forest is statistically very close (73.3% CV, 0.815 AUC), with lower variance across folds (±0.3% vs. ±0.5%).
3) The soft-voting ensemble (72.8% CV, 0.818 AUC) is within one standard deviation of the best single model, confirming that combining complementary learners does not degrade performance on this dataset.
4) Decision Trees underperform both gradient-based and ensemble methods, consistent with their known sensitivity to feature interactions at shallow depths.

B. Association statistics

Table IV reports the mean chi-square and phi coefficients computed over the five CV folds for the three best-performing models.

TABLE IV
Mean chi-square and phi coefficient (5 CV folds).

Model	χ^2 (mean)	ϕ (mean)
Logistic Reg	≈ 11,000	+0.21
Random Forest	≈ 10,800	+0.21
Soft-voting	≈ 10,600	+0.20

The phi coefficient of +0.21 for Logistic Regression and Random Forest indicates a weak-to-moderate positive association between the predicted and true class, which is consistent with the 73% accuracy on a 14%-positive dataset: a model that always predicts "no" would achieve 86% accuracy and $\phi \approx 0$, so the positive phi confirms genuine signal beyond the base rate.

C. Statistical test

A paired t-test on the five per-fold CV accuracy scores of Logistic Regression (mean = 0.7337, std = 0.0050) and Random Forest (mean = 0.7327, std = 0.0033) yields a p-value > 0.05, indicating that the difference between these two models is not statistically significant at the 95% confidence level. Both models should therefore be considered equivalent for practical purposes on this dataset.

V. Discussion

Class imbalance. The 86/14 positive/negative split means that accuracy alone is a misleading metric: a trivial always-negative classifier scores 86%. All models in this study therefore use class_weight="balanced", which up-weights the minority class during training. Under this setting the F1 scores (40–44%) are the more informative measure: a model that identifies approximately 40% of the actual diabetics while maintaining 73% overall accuracy is performing meaningfully above the base rate.

Why the ensemble does not win. In many tabular ML settings a soft-voting ensemble outperforms its individual components. In this study the ensemble (72.8%) is marginally below Logistic Regression (73.4%) and Random Forest (73.3%). A plausible explanation is that the three base learners—Logistic Regression, a shallow Decision Tree, and Random Forest—are not sufficiently diverse: the Decision Tree contributes the least signal, and its vote pulls the ensemble mean slightly away from the stronger learners' consensus. This is consistent with the finding of Arman et al. [8] that the benefit of voting depends critically on the diversity of the base estimators.

Leakage-free design. By placing the preprocessor inside each model's Pipeline, we ensure that StandardScaler parameters (mean and standard deviation) and the OneHotEncoder category lists are fitted only on the training portion of each CV fold. This eliminates the most common source of optimistic bias in tabular cross-validation [3].

Limitations. (i) The BRFSS data are self-reported; BMI, smoking status, and medical history are subject to recall and social-desirability bias. (ii) The SVM models with probability=True are computationally expensive at this scale; their full results are in the supplementary notebook. (iii) We do not perform hyperparameter optimisation; the reported results use reasonable fixed hyperparameters chosen for reproducibility. (iv) The dataset is from 2015; a model trained on it may not generalise to current population health patterns.

VI. Conclusion

We presented a systematic comparison of nine classifiers on the full 253,680-row CDC BRFSS 2015 binary diabetes dataset, all evaluated under a single leakage-free scikit-learn pipeline. Logistic Regression and Random Forest achieved the highest cross-validated accuracy (73.4% and 73.3% respectively) and are statistically indistinguishable from one another. A soft-voting ensemble of the three strongest base learners achieved comparable performance, confirming that ensemble benefits are dataset- and configuration-dependent rather than automatic. The chi-square and phi statistics reported here provide a distribution-free complement to accuracy and AUC for quantifying how strongly each model's predictions track the true class. The full reproducible notebook, including all nine models, confusion matrices, clustering analysis, and statistical tests, is available alongside this paper.

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